## Long division

**1**

a $\dfrac{z}{10}+1$

b $9x^2+5x^4$

c $6x^3-4x^2+8$

d $8x^3+2x^2-5x-7$

**2** $\dfrac{6x^7-8x^5}{2x^3}=3x^4-4x^2$

**3** $16x^8-10x^6+12x^4$

**4** $x^3+0x^2+0x+1$

**5**

a $28$

b $x-9$

**6**

a Quotient: $x-12$, Remainder: $58$

b Quotient: $x-10$, Remainder: $61$

c Quotient: $x+6$, Remainder: $20$

d Quotient: $2x^2+3x+4$, Remainder: $0$

e Quotient: $3x^2+2$, Remainder: $0$

f Quotient: $3x+2$, Remainder: $0$

g Quotient: $x^3-3x^2+9x-27$, Remainder: $0$

h Quotient: $4x^2+12x-5$, Remainder: $0$

---

## Synthetic division

**7** Yes

**8**

a $x^2-4x-5$

b $x^2+3x+6$

c $x^2+5x+2$

d $3x^2-4x+5+\dfrac{3}{x-5}$

e $-3x^2+12x+4+\dfrac{1}{x+4}$

f $-3x^2-3x+9+\dfrac{1}{x+3}$

g $x^3+3x^2+4x+2+\dfrac{1}{x-5}$

h $x^3-4x^2-5x$

i $2x^3+3x^2-2x$

j $x^3+4x^2-3x+2$

k $x^4-2x^3+4x^2-8x+16$

l $x^6+2x^5+4x^4+8x^3+16x^2+32x+64$

**9**

a $x^2+2x+4$

b $(x+4)\left(x^2+2x+4\right)$

**10**

a $x^2-3x-4+\dfrac{14}{x+2}$

b $8$

**11**

a

i It represents the remainder when the polynomial $P$ is divided by $x-5$.

ii $x+4$

iii $0$

b

i It represents the remainder when the polynomial $P$ is divided by $x-5$.

ii $3x^2-3x+4+\dfrac{4}{x-5}$

iii $4$

**12**

a It represents the remainder when the polynomial $P$ is divided by $x-0.5$.

b $x^2+0.5x-0.75+\dfrac{1.625}{x-0.5}$

c $1.625$

**13**

a It represents the remainder when the polynomial $P(x)$ is divided by $x-\sqrt{2}$.

b $x^3+\sqrt{2}x^2-4x-4\sqrt{2}+\dfrac{3}{x-\sqrt{2}}$

c $3$

**14**

a $x-2$

b $0$

**15**

a $x-4+\dfrac{1}{x-5}$

b $1$

**16**

a $x^4+x^3+x^2+x+1$

b $0$

**17** $4\sqrt[3]{3}+5$

**18**

a $3x^2+2x-4$

b Yes

c $3x^2+14x+100x+\dfrac{816}{x-8}$

d No

**19**

a $x^3-4x^2+2x-3$

b Yes

c $x^3-12x^2+58x-187x+\dfrac{576}{x+3}$

d No

**20**

a $x-4$

b Yes

**21**

a $x-2-\dfrac{1}{x+5}$

b No

**22**

a $3x^2-5x+3+\dfrac{1}{x+3}$

b No

**23**

a $3x^2+3x$

b Yes

**24**

a $2x^2+2x+2$

b Yes

**25**

a $8x^2+4x+4-\dfrac{1}{x-0.25}$

b No

**26**

a $3x^5+3\sqrt{2}x^4+5x^3+5\sqrt{2}x^2+12x+12\sqrt{2}$

b Yes

**27**

a $\pm 4,\ \pm 1,\ \pm 2,\ \pm \dfrac{1}{2}$

b No

c Yes

d $(2x+1)\left(x^2-x-4\right)=0$

e $\dfrac{1}{2}, \dfrac{1\pm\sqrt{17}}{2}$

**28**

a $\pm 1,\ \pm 2,\ \pm 3,\ \pm 6$

b No

c Yes

d $(x-1)(x-2)(x-3)$

e $x=1,2,3$

**29**

a $x=1$

b $6x^2-5x+1$

c Yes

d $x=1, \dfrac{1}{2}, \dfrac{1}{3}$

**30**

a $x=1$

b $x^2+4x+3$

c Yes

d $(x-1)(x+1)(x+3)$

e $x=1,\ -1,\ -3$

## Applications

**31**

a $\dfrac{0.2x^3+0.6x^2+0.45x+0.1}{x+0.5}$

b $0.2x^2+0.5x+0.2$ units

**32** $x^3-3x^2+2x-2$ in

**33** $\left(x^3-2\right)$ units

**34** $\left(3x^3+2x^2+5x+7\right)$ units

**35** $\left(x^2+x-4\right)$ units

**36** $\left(26n^2+22n+58\right)$ units

**37** $\left(2x^2+6x+1\right)$ units

**38** $\left(2x^2-4x+9\right)$ hours

**39** $\left(4x^4-3x^3+8x^2-6x+7\right)\ \mathrm{m}^2$

**40** $4$ days